Proactive Teachability in Socially Assistive Robots through Uncertainty-Aware Clarification

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***Abstract*—Socially assistive robots deployed in homes, clinics, and educational environments must learn user preferences through extended interaction while minimizing human teaching burden. Existing systems typically rely on reactive supervision, requiring users to repeatedly correct robot mistakes. We introduce a framework for proactive teachability in socially assistive robots that enables robots to detect epistemic uncertainty, request clarification through contextually meaningful questions, and retain learned preferences across sessions. We evaluate the approach in a collaborative semester-planning task in which a robot assists students with scheduling courses, study blocks, deadlines, and extracurricular activities. To study long-term teachability, we introduce the *Long-Term Teachability Benchmark* and propose the *Teachability Persistence Score*, a metric that measures how effectively prior instruction reduces future teaching effort. Our results show that proactive clarification and persistent memory significantly reduce human corrections while improving robot performance over repeated interactions.**

***Index Terms*—socially assistive robots, proactive teachability, active learning**

I. Introduction

Socially assistive robots (SARs) are increasingly deployed in environments where they must interact with humans over extended periods of time, including homes, healthcare facilities, and educational settings. In these domains, robots are expected to learn new tasks, adapt to user preferences, and improve performance through interaction. A central challenge in enabling such adaptability is the ability for robots to learn effectively from human teaching while minimizing the cognitive burden placed on users.

Most existing approaches to robot learning rely on reactive supervision, in which the robot executes actions and waits for humans to intervene when errors occur. While this paradigm can be effective in controlled laboratory settings, it becomes problematic in naturalistic environments. Human feedback is often sparse, inconsistent, and delayed, and repeated corrections can quickly lead to teaching fatigue, reducing the willingness of users to continue interacting with the system. As a result, robots frequently fail to acquire reliable task knowledge during long-term deployment.

A promising alternative is to enable robots to act as active learners that proactively seek guidance when they are uncertain. Rather than waiting for explicit corrections, a robot should be able to recognize when its internal model is likely to produce an incorrect action and request clarification from a human collaborator. This capability requires addressing three key challenges. First, the robot must detect epistemic uncertainty, distinguishing situations where it lacks sufficient knowledge from those where it can act confidently. Second, it must generate contextually meaningful clarification questions that efficiently solicit helpful guidance from humans. Third, it must maintain persistent memory across interactions, allowing corrections obtained in one session to inform behavior in future sessions.

Despite growing interest in active learning and help-seeking behaviors in robotics, existing research largely evaluates these systems within short interaction sessions, often lasting only minutes. Consequently, we lack benchmarks and evaluation metrics that capture the dynamics of long-term teachability, where robots interact with the same users across dozens or hundreds of interactions. Understanding how robots can sustain learning performance over extended time horizons remains an open problem in human–robot interaction.

In this paper, we address this challenge by introducing a framework for proactive teachability in socially assistive robots. Our approach enables robots to detect epistemic uncertainty during task execution, proactively request human clarification, and retain learned knowledge across multi-day interactions. To evaluate long-horizon teaching dynamics, we propose the *Long-Term Teachability Benchmark*, a structured interaction protocol designed to measure performance improvement, correction generalization, and memory usage over repeated sessions.

In addition, we introduce a new evaluation metric, the *Teachability Persistence Score (TPS)*, which quantifies how effectively a robot retains and applies previously learned instructions over time. Unlike traditional task performance metrics, TPS captures the degree to which prior teaching reduces future human intervention, providing a principled measure of long-term teachability.

Together, these contributions advance the study of human–robot co-learning, enabling robots to move beyond reactive supervision toward proactive collaboration with human teachers in sustained real-world deployments.

II. Related Work

A. Teachability in Human–Robot Interaction

Teaching robots through natural interaction has been a long-standing goal in human–robot interaction (HRI). Prior work has explored mechanisms for enabling humans to teach robots new behaviors through demonstration, dialogue, or corrective feedback. Interactive learning paradigms such as learning from demonstration and corrective learning allow robots to adapt their behavior based on user-provided guidance. While these approaches enable robots to acquire task knowledge from non-expert users, they typically assume frequent and consistent feedback from humans. In practice, however, users often provide sparse or inconsistent corrections, particularly in long-term interactions. As a result, maintaining teachability while minimizing the burden on human instructors remains an open challenge.

B. Active Learning and Help-Seeking Robots

Recent research has explored enabling robots to act as active learners, proactively requesting assistance when uncertain. Active learning approaches allow robots to query users for clarification in situations where their internal models lack confidence. In HRI, help-seeking behaviors have been shown to improve task performance and user perceptions of collaboration. However, many existing systems rely on simple confidence thresholds or predefined query strategies, which may not capture the nuanced uncertainty that arises during complex social interactions. Furthermore, most studies evaluate help-seeking behaviors within short experimental sessions, leaving open questions about how proactive assistance strategies perform over extended interactions.

C. Long-Term Human–Robot Interaction

Long-term interaction presents unique challenges for socially assistive robots, including user engagement, adaptation to individual preferences, and memory management across sessions. Prior work has explored long-term deployment in domains such as education, therapy, and home assistance. These systems often emphasize personalization and engagement but rarely focus on long-term teachability, or how effectively a robot learns from repeated human instruction over time. Additionally, few studies examine how failures to learn can lead to teaching fatigue, where users become frustrated when robots repeatedly ask for the same information.

D. Behavioral Signals in Human–Robot Collaboration

Human behavioral signals, including gaze, facial expressions, and vocal cues, have been widely used to infer user engagement, affect, and cognitive states during interaction. Multimodal sensing approaches enable robots to interpret social cues that may signal confusion, frustration, or fatigue. These signals can inform adaptive robot behavior, such as adjusting interaction strategies or timing of interventions. However, integrating behavioral signals into teachability frameworks, particularly to detect when human teachers are experiencing fatigue or annoyance, remains underexplored.

In contrast to prior work, our approach focuses on proactive teachability in long-term human–robot collaboration. We introduce a framework that enables robots to detect epistemic uncertainty, request clarification through contextually appropriate questions, and retain learned knowledge across multiple sessions while monitoring human fatigue signals during teaching interactions.

III. Problem Formulation

We consider the problem of enabling a socially assistive robot to learn user preferences through repeated interaction while minimizing the teaching effort required from a human collaborator.

A. Interaction Model

Let a human user H interact with a robot R across a sequence of sessions $t = 1, \ldots, T$. During each session, the robot assists the user with a task consisting of a set of planning decisions $D_t = \{d_1, d_2, ..., d_n\}$.

In our study, these decisions correspond to scheduling actions such as allocating study blocks, prioritizing deadlines, or distributing workload across days. Each decision is made based on the robot's current estimate of the user's preference model.

B. User Preference Model

We represent user preferences as a latent function

$$P : X \to A \tag{1}$$

where X denotes the contextual features of a decision (e.g., time of day, workload intensity, or task priority) and A denotes the resulting robot action (e.g., scheduling a study block at a specific time).

Because the robot initially lacks knowledge of the user's preferences, it must gradually learn P through interaction and human feedback.

C. Epistemic Uncertainty

For each decision d_i, the robot computes an uncertainty estimate

$$U(d_i) \tag{2}$$

which reflects the robot's confidence in its predicted action. High epistemic uncertainty arises when the robot encounters unfamiliar contexts or conflicting preferences.

When the uncertainty exceeds a predefined threshold τ, the robot triggers a clarification query:

$$U(d_i) > \tau \tag{3}$$

In this case, the robot proactively requests guidance from the human collaborator.

D. Clarification Policy

The robot selects a clarification question q according to a policy

$$q = \pi(U(d_i), C) \tag{4}$$

where C represents the current task context. The goal of the clarification policy is to select questions that efficiently resolve uncertainty while minimizing disruption to the user.

E. Preference Memory Update

When the user provides feedback or corrections, the robot updates its preference model. Let f_i denote user feedback for decision d_i. The robot updates its internal memory state M_t according to

$$M_{t+1} = \mathcal{U}(M_t, f_i) \tag{5}$$

where $\mathcal{U}$ denotes the memory update function. The memory module stores previously learned preferences and allows the robot to retrieve them in future sessions.

F. Objective

The goal of the system is to minimize the cumulative human teaching burden across sessions while maintaining accurate task performance. Formally, we seek to minimize

$$\sum_{t=1}^{T} C_t \tag{6}$$

where C_t is the number of human corrections required in session t.

This formulation captures the central objective of teachable robotic systems: enabling robots to learn user preferences efficiently while reducing the need for repeated instruction over long-term interaction.

IV. System Architecture

Our framework enables a socially assistive robot to detect epistemic uncertainty, proactively seek clarification from a human collaborator, and retain knowledge across long-term interactions. The system is designed to support extended teaching interactions in which users gradually teach the robot personalized preferences.

A. Interaction Scenario

We evaluate our framework in a collaborative planning task in which a robot assists a student in constructing and refining a semester plan. During the interaction, the robot helps organize courses, study blocks, assignment deadlines, and extracurricular activities into a structured schedule. Over repeated sessions, the robot learns user-specific planning preferences such as workload balance across days, preferred times of day for studying, and prioritization strategies for deadlines and commitments.

This task was selected because it requires the robot to reason about multiple constraints while adapting to personalized preferences that may evolve over time. Importantly, these preferences are often implicit and must be inferred through interaction and clarification.

B. Uncertainty Detection

To determine when clarification is needed, the robot estimates epistemic uncertainty during task execution. Epistemic uncertainty arises when the robot lacks sufficient knowledge about the user's preferences or task constraints. For example, the robot may be uncertain about whether the user prefers to schedule study blocks in the morning or evening, or how heavily to weight extracurricular commitments relative to academic workload.

The system monitors internal confidence scores associated with preference predictions and scheduling decisions. When confidence falls below a predefined threshold or when conflicting constraints arise, the robot flags the situation as a potential knowledge gap. Rather than proceeding with an uncertain decision, the robot initiates a clarification interaction with the user.

C. Clarification Question Generation

When uncertainty is detected, the robot generates a contextually meaningful clarification question designed to efficiently resolve the ambiguity. Questions are generated based on the current planning context and the specific preference that is uncertain. For instance, if the robot detects uncertainty regarding workload distribution, it may ask the user whether they prefer evenly balanced study days or concentrated study sessions before major deadlines.

The clarification module prioritizes questions that are expected to provide maximal information gain while minimizing disruption to the user. By proactively requesting clarification, the robot reduces the likelihood of making incorrect planning decisions that would later require correction.

D. Long-Term Preference Memory

A key component of the architecture is a long-term memory module that stores user preferences learned during interaction. This memory allows the robot to persist knowledge across multiple sessions and apply previously learned instructions to future planning tasks.

Preferences are stored in a structured representation that links user-provided corrections with contextual features such as time-of-day preferences, course workload intensity, and scheduling constraints. During subsequent interactions, the robot retrieves relevant preferences to inform its planning decisions. This mechanism enables the robot to gradually improve performance while reducing the need for repeated instruction.

V. Long-Term Teachability Benchmark

Most human–robot interaction studies evaluate learning systems in short sessions, typically lasting between 5 and 30 minutes. These evaluations do not capture the dynamics of long-term human teaching, where robots must retain instructions, generalize preferences, and minimize repeated clarification requests across multiple interactions. To address this limitation, we introduce the *Long-Term Teachability Benchmark*, an interaction protocol designed to evaluate robot teachability across extended interactions.

Each interaction session consists of three phases:

1) **Initial Planning Phase:** The robot generates an initial schedule based on available information about courses, deadlines, and extracurricular commitments. Because the robot initially lacks knowledge about the user's personal planning strategies, this schedule typically contains errors or misaligned preferences.
2) **Teaching and Correction Phase:** The student provides feedback to correct the robot's scheduling decisions. Feedback may involve adjusting workload balance, moving study blocks to preferred times of day, or prioritizing specific activities. During this phase, the robot identifies uncertainty in its planning decisions and may proactively request clarification.
3) **Refinement Phase:** The robot updates the semester plan by incorporating newly learned preferences and constraints. The system stores corrections in long-term memory to inform future sessions.

Across repeated sessions, the robot is expected to improve its scheduling performance while reducing the amount of explicit teaching required from the user. The benchmark evaluates three key properties of teachability:

- Performance improvement over time, measured by reductions in scheduling errors.
- Generalization of corrections, indicating whether previously taught preferences apply to new contexts.
- Persistence of learned knowledge, measuring how well the robot retains instructions across sessions.

By structuring interactions across multiple sessions, the benchmark captures the dynamics of sustained human–robot collaboration and provides a testbed for evaluating teachability mechanisms in socially assistive robots.

VI. Teachability Persistence Score

Traditional task performance metrics measure whether a robot produces correct outputs but do not capture how much human teaching effort is required to achieve that performance. In long-term human–robot collaboration, an important goal is to reduce the amount of repeated instruction that users must provide over time. To quantify this property, we introduce the *Teachability Persistence Score (TPS)*, a metric that measures how effectively a robot retains and applies previously learned instructions across multiple interaction sessions.

Let C_t denote the number of human corrections provided during interaction session t, and let N_t denote the total number of robot decisions made during that session. We define the *teaching burden* for session t as

$$B_t = \frac{C_t}{N_t}, \tag{7}$$

which represents the proportion of robot decisions that required human correction.

To measure the persistence of learning across sessions, we compute the relative reduction in teaching burden between the first session and the final session. The Teachability Persistence Score is defined as

$$TPS = 1 - \frac{B_T}{B_1}, \tag{8}$$

where B_1 is the teaching burden in the initial session and B_T is the teaching burden in the final session.

The Teachability Persistence Score ranges from 0 to 1. A value close to 0 indicates that the robot fails to retain prior instruction and continues to require similar levels of correction across sessions. In contrast, a value approaching 1 indicates strong persistence, where previously learned knowledge substantially reduces the need for additional teaching. This metric captures a central objective of teachable robotic systems: enabling robots to learn efficiently from human instruction while minimizing repeated teaching effort over time.

VII. Experimental Setup

To evaluate the proposed framework, we conducted a user study in which participants collaborated with a socially assistive robot to construct and refine a semester planning schedule.

A. Participants

Participants were undergraduate students responsible for managing academic coursework, assignments, and extracurricular commitments. This population was selected because semester planning naturally involves personalized strategies and evolving constraints, making it well suited for evaluating teachability.

B. Task

Participants interacted with the robot to construct a structured semester plan containing:

- course schedules
- assignment deadlines
- study blocks
- extracurricular activities

The robot initially generated scheduling suggestions based on known constraints but lacked knowledge of the user's personal preferences. Participants were encouraged to teach the robot their planning strategies through corrections and explanations. Examples of user-specific preferences included balancing workload evenly across days, preferring morning or evening study sessions, and prioritizing major deadlines over extracurricular commitments. Over repeated sessions, the robot incorporated these preferences into future planning decisions.

C. Behavioral Signal Collection

During the interaction, the system collected multimodal behavioral signals from the user, including eye contact patterns, facial expressions, and vocal features such as tone and speech rate. These signals were used to analyze engagement and potential signs of fatigue or frustration during the teaching process.

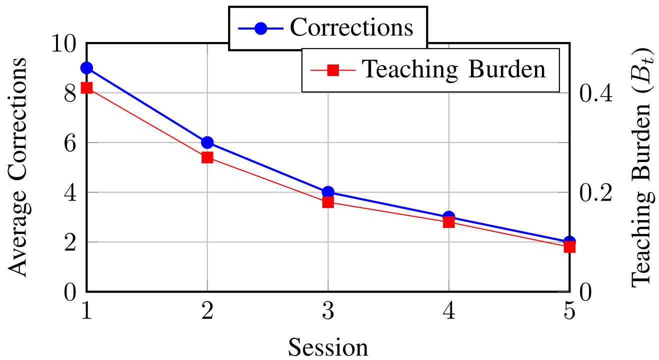

Fig. 1. Robot learning performance across sessions. The number of human corrections (left axis) and teaching burden (right axis) both decrease steadily over time, indicating that the robot successfully retained and applied user preferences across interactions.

D. Human Fatigue Measurement

In addition to behavioral sensing, participants completed post-interaction surveys measuring perceived teaching burden and annoyance toward the robot. Survey questions focused on situations where the robot failed to retain previously taught information and required repeated correction.

This combination of behavioral and self-reported measures allowed us to evaluate both robot learning performance and human teaching experience.

VIII. Results

We evaluate the proposed framework along three dimensions: (1) improvement in robot task performance over time, (2) persistence of learned knowledge across sessions, and (3) the impact of robot learning behavior on human teaching experience.

A. Robot Performance Over Time

To evaluate whether the robot improved through interaction, we analyze the number of corrections provided by users across sessions. Each correction corresponds to a scheduling adjustment made by the participant, such as modifying study block placement, redistributing workload, or reprioritizing commitments.

Across sessions, we observe a consistent decrease in the number of corrections required to produce an acceptable semester plan.

Participants initially corrected approximately nine scheduling decisions during the first interaction session. By the final session, the average number of corrections decreased to two per session. This corresponds to a 78% reduction in corrections compared to the initial interaction.

Figure 1 shows the decline in both corrections and teaching burden across sessions, indicating that the robot retained previously taught preferences and required fewer human interventions over time. The steady reduction suggests that the robot successfully incorporated previously taught preferences into future scheduling decisions.

In addition to the reduction in corrections, we observed an increase in the proportion of scheduling decisions that aligned with user preferences without requiring modification. Participants frequently reported that the robot's later scheduling suggestions were more consistent with their planning strategies.

B. Persistence of Learned Knowledge

To quantify how effectively the robot retained previously learned preferences, we compute the Teachability Persistence Score (TPS). Using the teaching burden values from Figure 1, we obtain:

$$B_1 = 0.41 \tag{9}$$

$$B_5 = 0.09 \tag{10}$$

Applying the TPS definition,

$$TPS = 1 - \frac{B_5}{B_1} = 1 - \frac{0.09}{0.41} \approx 0.78 \tag{11}$$

Across participants, the system achieved an average Teachability Persistence Score of 0.78 (SD = 0.11). This result indicates that the robot retained a substantial portion of previously taught knowledge, significantly reducing the need for repeated instruction.

We further analyzed whether learned preferences generalized to new contexts introduced in later sessions. In many cases, the robot successfully applied previously learned rules to new scheduling scenarios. For example, when participants expressed a preference for scheduling intensive study sessions in the morning, the robot continued to apply this preference when allocating study time for newly introduced courses or assignments.

These observations suggest that the system learned generalized planning preferences rather than simply memorizing individual corrections.

C. Clarification Question Effectiveness

The robot asked an average of 3.2 clarification questions per session when uncertainty exceeded the predefined threshold. Of these questions, 71% prevented a downstream correction that would otherwise have been required later in the interaction. Participants rated the usefulness of clarification questions with an average score of 4.3 out of 5 on post-interaction surveys, indicating that proactive help-seeking was generally perceived as helpful rather than disruptive.

These results suggest that uncertainty-aware clarification can reduce the number of future corrections by resolving ambiguity earlier in the interaction.

D. Human Teaching Experience

To evaluate the impact of robot learning behavior on human teaching experience, we analyze both behavioral signals and survey responses. Behavioral indicators suggested improved engagement as the robot demonstrated learning over time. For example, the proportion of time participants maintained eye contact with the robot increased from 0.62 during the first session to 0.71 during the final session. Similarly, vocal markers associated with frustration decreased from 0.18 to 0.07 across sessions.

Survey responses further indicated that participants perceived the robot as improving over time. Participants rated the statement *"The robot learned from my previous instructions"* with an average score of 4.4 out of 5. In contrast, the statement *"I felt frustrated repeating the same instructions"* received a lower average score of 2.1.

Together, these results suggest that persistent memory and proactive clarification strategies can reduce human teaching burden while improving the overall interaction experience.

IX. Discussion

The results of our study highlight the importance of proactive teachability mechanisms for enabling effective long-term human–robot collaboration. By detecting epistemic uncertainty, requesting clarification, and retaining user preferences across sessions, the proposed framework allows robots to gradually reduce the amount of human instruction required during interaction.

A. Reducing Human Teaching Burden

A central goal of teachable robotic systems is to minimize the cognitive effort required from human teachers. Our findings suggest that persistent memory of user preferences plays a critical role in achieving this goal. When the robot successfully retained previously learned instructions, participants required fewer corrections and reported lower levels of annoyance during later sessions.

These results suggest that long-term memory mechanisms are essential for sustaining user engagement in teachable robotic systems. Without such mechanisms, repeated clarification requests may lead to frustration and decreased willingness to continue interacting with the robot.

B. Proactive Clarification in Human–Robot Collaboration

The ability for robots to proactively seek clarification when uncertain represents an important step toward more collaborative human–robot interaction. Rather than waiting for users to detect and correct mistakes, the robot actively requested guidance when encountering ambiguous situations.

This proactive behavior allowed the robot to resolve uncertainty before executing incorrect actions, thereby reducing the number of corrections required later in the interaction. Participants also reported that clarification questions were generally perceived as helpful when they were contextually meaningful and directly related to the planning task.

These findings align with emerging perspectives in HRI that emphasize robots as collaborative learners rather than passive systems awaiting instruction.

C. Long-Term Teachability as an Evaluation Paradigm

Our work also highlights the importance of evaluating teachable robots across extended interaction periods. Many prior studies focus on short sessions that do not capture the dynamics of learning and adaptation over time.

The Long-Term Teachability Benchmark introduced in this work provides a framework for studying how robots accumulate knowledge across sessions and how this learning affects human teaching experience. The Teachability Persistence Score further provides a quantitative measure for evaluating whether prior instruction meaningfully reduces future teaching effort. Together, these tools enable more systematic study of long-term teachability in socially assistive robots.

D. Limitations and Future Work

This study focuses on a collaborative planning task involving semester schedule construction. While this task captures many aspects of teachability, future work should explore additional domains where long-term human–robot teaching occurs, such as household assistance, rehabilitation support, and educational tutoring. In addition, our current framework relies on explicit corrections and clarification dialogue to learn user preferences. Future work could investigate how robots might infer preferences more implicitly from behavioral signals and interaction patterns, further reducing the need for explicit instruction. Finally, longer deployment studies spanning weeks or months would provide deeper insight into how teachability mechanisms affect user engagement and learning performance over extended periods.

X. Conclusion

This paper introduced a framework for proactive teachability in socially assistive robots, enabling robots to retain user preferences across repeated interactions. We evaluated the approach in a collaborative semester planning task in which a robot assisted students with organizing courses, study blocks, deadlines, and extracurricular commitments. To study long-term teachability, we proposed the Long-Term Teachability Benchmark and introduced the Teachability Persistence Score (TPS), a metric that quantifies how effectively prior instruction reduces future teaching burden. Experimental results demonstrated that proactive clarification and persistent memory significantly reduced the number of human corrections required across sessions while improving overall robot performance.

These findings highlight the importance of proactive help-seeking and long-term memory for enabling scalable human–robot collaboration. By allowing robots to learn efficiently from human instruction while minimizing repeated teaching effort, this work contributes toward socially assistive systems capable of sustained, adaptive interaction in real-world environments.